# Zener Voltage Regulators

## 250 mW SOT−23 Surface Mount

# BZX84CxxxET1G Series, SZBZX84CxxxET1G Series

This series of Zener diodes is offered in the convenient, surface mount plastic SOT−23 package. These devices are designed to provide voltage regulation with minimum space requirement. They are well suited for applications such as cellular phones, hand held portables, and high density PC boards.

### Specification Features

- 250 mW Rating on FR−4 or FR−5 Board
- Zener Breakdown Voltage Range − 2.4 V to 75 V
- Package Designed for Optimal Automated Board Assembly
- Small Package Size for High Density Applications
- ESD Rating of Class 3 (> 16 kV) per Human Body Model
- Peak Power − 225 W (8 X 20 μs)
- SZ Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC−Q101 Qualified and PPAP Capable
- These Devices are Pb−Free and are RoHS Compliant

### Mechanical Characteristics

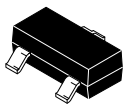
**CASE:** Void-free, transfer-molded, thermosetting plastic case
**FINISH:** Corrosion resistant finish, easily solderable
**MAXIMUM CASE TEMPERATURE FOR SOLDERING PURPOSES:** 260°C for 10 Seconds
**POLARITY:** Cathode indicated by polarity band
**FLAMMABILITY RATING:** UL 94 V−0

### MAXIMUM RATINGS

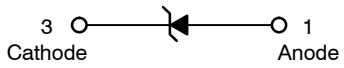
| Rating | Symbol | Max | Unit |
|---|---|---|---|
| Peak Power Dissipation @ 20 μs (Note 1) @ $T_L \le 25°C$ | $P_{pk}$ | 225 | W |
| Total Power Dissipation on FR−5 Board, (Note 2) @ $T_A = 25°C$<br>Derated above 25°C<br>Thermal Resistance, Junction−to−Ambient | $P_D$<br><br>$R_{\theta JA}$ | 250<br>2.0<br>500 | mW<br>mW/°C<br>°C/W |
| Total Power Dissipation on Alumina Substrate, (Note 3) @ $T_A = 25°C$<br>Derated above 25°C<br>Thermal Resistance, Junction−to−Ambient | $P_D$<br><br>$R_{\theta JA}$ | 300<br>2.4<br>417 | mW<br>mW/°C<br>°C/W |
| Junction and Storage Temperature Range | $T_J$, $T_{stg}$ | −65 to +150 | °C |

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

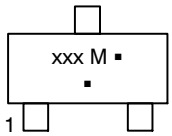
1. Nonrepetitive current pulse per Figure 9.
2. FR−5 = 1.0 X 0.75 X 0.62 in.
3. Alumina = 0.4 X 0.3 X 0.024 in, 99.5% alumina.



**SOT−23**
**CASE 318**
**STYLE 8**




### MARKING DIAGRAM




xxx = Device Code
M = Date Code*
▪ = Pb−Free Package

(Note: Microdot may be in either location)

*Date Code orientation may vary depending upon manufacturing location.

### ORDERING INFORMATION

| Device | Package | Shipping† |
|---|---|---|
| BZX84CxxxET1G | SOT−23 (Pb−Free) | 3,000 / Tape & Reel |
| SZBZX84CxxxET1G | SOT−23 (Pb−Free) | 3,000 / Tape & Reel |
| BZX84CxxxET3G | SOT−23 (Pb−Free) | 10,000 / Tape & Reel |
| SZBZX84CxxxET3G | SOT−23 (Pb−Free) | 10,000 / Tape & Reel |

NOTE: Some of the devices on this data sheet have been **DISCONTINUED**. Please refer to the table on page 2.

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, BRD8011/D.

### DEVICE MARKING INFORMATION

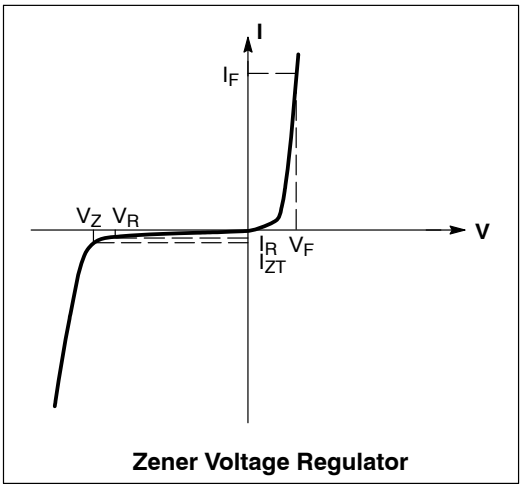
See specific marking information in the device marking column of the Electrical Characteristics table on page 2 of this data sheet.

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## ELECTRICAL CHARACTERISTICS

(Pinout: 1-Anode, 2-No Connection, 3-Cathode) ($T_A$ = 25°C unless otherwise noted, $V_F$ = 0.90 V Max. @ $I_F$ = 10 mA)

| Symbol | Parameter |
|---|---|
| $V_Z$ | Reverse Zener Voltage @ $I_{ZT}$ |
| $I_{ZT}$ | Reverse Current |
| $Z_{ZT}$ | Maximum Zener Impedance @ $I_{ZT}$ |
| $I_R$ | Reverse Leakage Current @ $V_R$ |
| $V_R$ | Reverse Voltage |
| $I_F$ | Forward Current |
| $V_F$ | Forward Voltage @ $I_F$ |
| $\Theta V_Z$ | Maximum Temperature Coefficient of $V_Z$ |
| C | Max. Capacitance @ $V_R$ = 0 and f = 1 MHz |




**Zener Voltage Regulator**

## ELECTRICAL CHARACTERISTICS

(Pinout: 1-Anode, 2-No Connection, 3-Cathode) ($T_A$ = 25°C unless otherwise noted, $V_F$ = 0.90 V Max. @ $I_F$ = 10 mA)

| Device* | Device Marking | $V_{Z1}$ (V) @ $I_{ZT1}$ = 5 mA (Note 4) | | | $Z_{ZT1}$ (Ω) @ $I_{ZT1}$ = 5 mA | $V_{Z2}$ (V) @ $I_{ZT2}$ = 1 mA (Note 4) | | $Z_{ZT2}$ (Ω) @ $I_{ZT2}$ = 1 mA | $V_{Z3}$ (V) @ $I_{ZT3}$=20 m-A (Note 4) | | $Z_{ZT3}$ (Ω) @ $I_{ZT3}$= 20 mA | Max Reverse Leakage Current | | $\theta_{VZ}$ (mV/k) @ $I_{ZT1}$=5 mA | | C (pF) @ $V_R$ = 0 f = 1 MHz |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Min | Nom | Max | | Min | Max | | Min | Max | | $I_R$ µA @ | $V_R$ (V) | Min | Max | |
| BZX84C2V4ET1G | BA1 | 2.2 | 2.4 | 2.6 | 100 | 1.7 | 2.1 | 600 | 2.6 | 3.2 | 50 | 50 | 1.0 | −3.5 | 0 | 450 |
| BZX84C6V2ET1G | BB3 | 5.8 | 6.2 | 6.6 | 10 | 5.6 | 6.6 | 150 | 5.8 | 6.8 | 6 | 3.0 | 4.0 | 0.4 | 3.7 | 185 |

**DISCONTINUED** (Note 5)

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BZX84C2V7ET1G | BA2 | 2.5 | 2.7 | 2.9 | 100 | 1.9 | 2.4 | 600 | 3.0 | 3.6 | 50 | 20 | 1.0 | −3.5 | 0 | 450 |
| BZX84C13ET1G | BC2 | 12.4 | 13 | 14.1 | 30 | 12.3 | 14 | 170 | 12.5 | 14.2 | 15 | 0.1 | 8.0 | 7.0 | 11 | 120 |
| BZX84C16ET1G | BC4 | 15.3 | 16 | 17.1 | 40 | 15.2 | 17 | 200 | 15.4 | 17.2 | 20 | 0.05 | 11.2 | 10.4 | 14 | 105 |
| BZX84C22ET1G | BC7 | 20.8 | 22 | 23.3 | 55 | 20.7 | 23.2 | 250 | 20.9 | 23.4 | 25 | 0.05 | 15.4 | 16.4 | 20 | 85 |
| BZX84C10ET1G | BB8 | 9.4 | 10 | 10.6 | 20 | 9.3 | 10.6 | 150 | 9.4 | 10.7 | 10 | 0.2 | 7.0 | 4.5 | 8.0 | 130 |
| BZX84C7V5ET1G | BB5 | 7.0 | 7.5 | 7.9 | 15 | 6.9 | 7.9 | 80 | 7.0 | 8.0 | 6 | 1.0 | 5.0 | 2.5 | 5.3 | 140 |
| BZX84C5V1ET1G | BB1 | 4.8 | 5.1 | 5.4 | 60 | 4.2 | 5.3 | 480 | 5.0 | 5.9 | 15 | 2.0 | 2.0 | −2.7 | 1.2 | 225 |
| BZX84C4V3ET1G | BA7 | 4.0 | 4.3 | 4.6 | 90 | 3.3 | 4.0 | 600 | 4.4 | 5.1 | 30 | 3.0 | 1.0 | −3.5 | 0 | 450 |
| BZX84C6V8ET1G | BB4 | 6.4 | 6.8 | 7.2 | 15 | 6.3 | 7.2 | 80 | 6.4 | 7.4 | 6 | 2.0 | 4.0 | 1.2 | 4.5 | 155 |
| BZX84C3V3ET1G | BA4 | 3.1 | 3.3 | 3.5 | 95 | 2.3 | 2.9 | 600 | 3.6 | 4.2 | 40 | 5.0 | 1.0 | −3.5 | 0 | 450 |
| BZX84C18ET1G | BC5 | 16.8 | 18 | 19.1 | 45 | 16.7 | 19 | 225 | 16.9 | 19.2 | 20 | 0.05 | 12.6 | 12.4 | 16 | 100 |
| BZX84C5V6ET1G | BB2 | 5.2 | 5.6 | 6.0 | 40 | 4.8 | 6.0 | 400 | 5.2 | 6.3 | 10 | 1.0 | 2.0 | −2 | 2.5 | 200 |
| BZX84C3V0ET1G | BA3 | 2.8 | 3.0 | 3.2 | 95 | 2.1 | 2.7 | 600 | 3.3 | 3.9 | 50 | 10 | 1.0 | −3.5 | 0 | 450 |
| BZX84C8V2ET1G | BB6 | 7.7 | 8.2 | 8.7 | 15 | 7.6 | 8.7 | 80 | 7.7 | 8.8 | 6 | 0.7 | 5.0 | 3.2 | 6.2 | 135 |
| BZX84C15ET1G | BC3 | 13.8 | 15 | 15.6 | 30 | 13.7 | 15.5 | 200 | 13.9 | 15.7 | 20 | 0.05 | 10.5 | 9.2 | 13 | 110 |
| BZX84C11ET1G | BB9 | 10.4 | 11 | 11.6 | 20 | 10.2 | 11.6 | 150 | 10.4 | 11.8 | 10 | 0.1 | 8.0 | 5.4 | 9.0 | 130 |
| BZX84C20ET1G | BC6 | 18.8 | 20 | 21.2 | 55 | 18.7 | 21.1 | 225 | 18.9 | 21.4 | 20 | 0.05 | 14 | 14.4 | 18 | 85 |
| BZX84C24ET1G | BC8 | 22.8 | 24 | 25.6 | 70 | 22.7 | 25.5 | 250 | 22.9 | 25.7 | 25 | 0.05 | 16.8 | 18.4 | 22 | 80 |
| BZX84C3V6ET1G | BA5 | 3.4 | 3.6 | 3.8 | 90 | 2.7 | 3.3 | 600 | 3.9 | 4.5 | 40 | 5.0 | 1.0 | −3.5 | 0 | 450 |
| BZX84C4V7ET1G | BA9 | 4.4 | 4.7 | 5.0 | 80 | 3.7 | 4.7 | 500 | 4.5 | 5.4 | 15 | 3.0 | 2.0 | −3.5 | 0.2 | 260 |
| BZX84C9V1ET1G | BB7 | 8.5 | 9.1 | 9.6 | 15 | 8.4 | 9.6 | 100 | 8.5 | 9.7 | 8 | 0.5 | 6.0 | 3.8 | 7.0 | 130 |
| BZX84C3V9ET1G | BA6 | 3.7 | 3.9 | 4.1 | 90 | 2.9 | 3.5 | 600 | 4.1 | 4.7 | 30 | 3.0 | 1.0 | −3.5 | −2.5 | 450 |



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## ELECTRICAL CHARACTERISTICS

(Pinout: 1-Anode, 2-No Connection, 3-Cathode) ($T_A$ = 25°C unless otherwise noted, $V_F$ = 0.90 V Max. @ $I_F$ = 10 mA)

| Device* | Device Marking | $V_{Z1}$ Below @ $I_{ZT1}$ = 2 mA | | | $Z_{ZT1}$ Below @ $I_{ZT1}$ = 2 mA | $V_{Z2}$ Below @ $I_{ZT2}$ = 0.1 mA | | $Z_{ZT2}$ Below @ $I_{ZT4}$ = 0.5 mA | $V_{Z3}$ Below @ $I_{ZT3}$ = 10 mA | | $Z_{ZT3}$ Below @ $I_{ZT3}$ = 10 mA | Max Reverse Leakage Current | | $\theta_{VZ}$ (mV/k) Below @ $I_{ZT1}$ = 2 mA | | C (pF) @ $V_R$ = 0 f = 1 MHz |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Min | Nom | Max | | Min | Max | | Min | Max | | $I_R$ µA @ | $V_R$ (V) | Min | Max | |
| BZX84C30ET1G | BD1 | 28 | 30 | 32 | 80 | 27.8 | 32 | 300 | 28.1 | 32.4 | 50 | 0.05 | 21 | 24.4 | 29.4 | 70 |
| **DISCONTINUED** (Note 5) | | | | | | | | | | | | | | | | |
| BZX84C47ET1G | BD5 | 44 | 47 | 50 | 170 | 43.7 | 50 | 375 | 44.1 | 50.5 | 90 | 0.05 | 32.9 | 42 | 51.8 | 40 |
| BZX84C56ET1G | BD7 | 52 | 56 | 60 | 200 | 51.5 | 60 | 425 | 52.1 | 60.8 | 110 | 0.05 | 39.2 | 52.2 | 63.8 | 40 |
| BZX84C62ET1G | BD8 | 58 | 62 | 66 | 215 | 57.4 | 66 | 450 | 58.2 | 67 | 120 | 0.05 | 43.4 | 58.8 | 71.6 | 35 |
| BZX84C43ET1G | BK6 | 40 | 43 | 46 | 150 | 39.7 | 46 | 375 | 40.1 | 46.5 | 80 | 0.05 | 30.1 | 37.6 | 46.6 | 40 |
| BZX84C36ET1G | BD3 | 34 | 36 | 38 | 90 | 33.8 | 38 | 350 | 34.1 | 38.4 | 60 | 0.05 | 25.2 | 30.4 | 37.4 | 70 |
| BZX84C27ET1G | BC9 | 25.1 | 27 | 28.9 | 80 | 25 | 28.9 | 300 | 25.2 | 29.3 | 45 | 0.05 | 18.9 | 21.4 | 25.3 | 70 |
| BZX84C39ET1G | BD4 | 37 | 39 | 41 | 130 | 36.7 | 41 | 350 | 37.1 | 41.5 | 70 | 0.05 | 27.3 | 33.4 | 41.2 | 45 |
| BZX84C33ET1G | BD2 | 31 | 33 | 35 | 80 | 30.8 | 35 | 325 | 31.1 | 35.4 | 55 | 0.05 | 23.1 | 27.4 | 33.4 | 70 |
| BZX84C12ET1G | BC1 | 11.4 | 12 | 12.7 | 25 | 11.2 | 12.7 | 150 | 11.4 | 12.9 | 10 | 0.1 | 8.0 | 6.0 | 10 | 130 |
| BZX84C51ET1G | BD6 | 48 | 51 | 54 | 180 | 47.6 | 54 | 400 | 48.1 | 54.6 | 100 | 0.05 | 35.7 | 46.6 | 57.2 | 40 |
| BZX84C75ET1G | BE1 | 70 | 75 | 79 | 255 | 69.4 | 79 | 500 | 70.3 | 80.2 | 140 | 0.05 | 52.5 | 73.4 | 88.6 | 35 |
| BZX84C68ET1G | BD9 | 64 | 68 | 72 | 240 | 63.4 | 72 | 475 | 64.2 | 73.2 | 130 | 0.05 | 47.6 | 65.6 | 79.8 | 35 |

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

* Include SZ-prefix devices where applicable.

4. Zener voltage is measured with a pulse test current $I_Z$ at an ambient temperature of 25°C
5. **DISCONTINUED:** These devices are not recommended for new design. Please contact your **onsemi** representative for information. The most current information on these devices may be available on www.onsemi.com.



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## TYPICAL CHARACTERISTICS

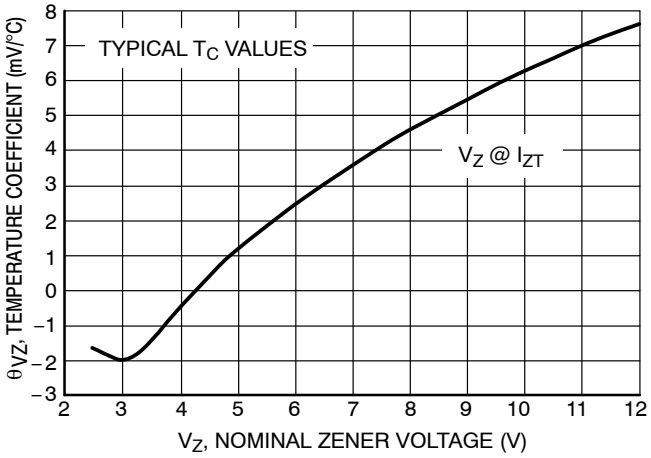



**Figure 1. Temperature Coefficients (Temperature Range −55°C to +150°C)**

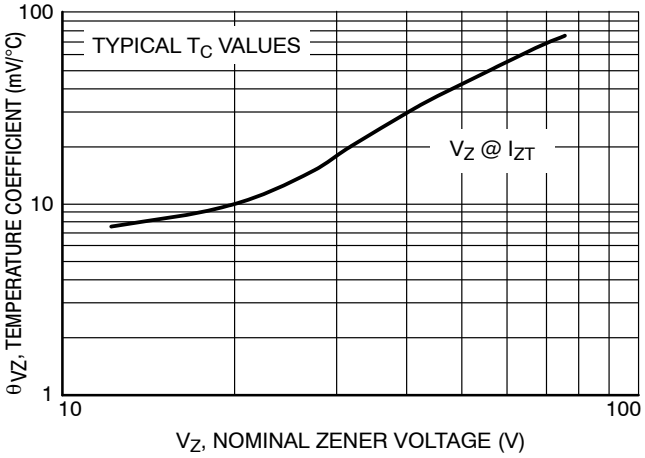



**Figure 2. Temperature Coefficients (Temperature Range −55°C to +150°C)**

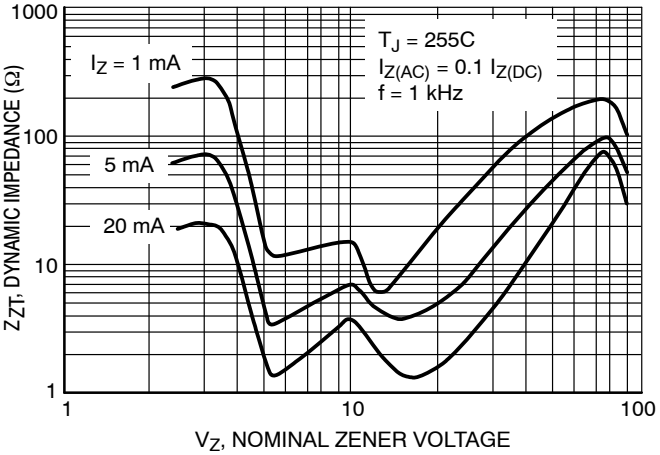



**Figure 3. Effect of Zener Voltage on Zener Impedance**

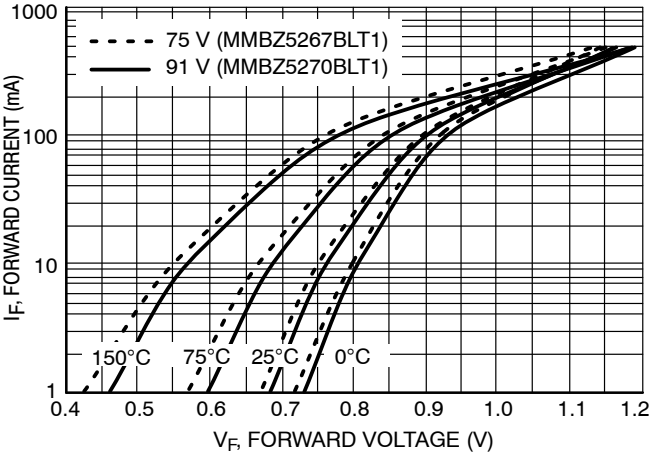



**Figure 4. Typical Forward Voltage**



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## TYPICAL CHARACTERISTICS

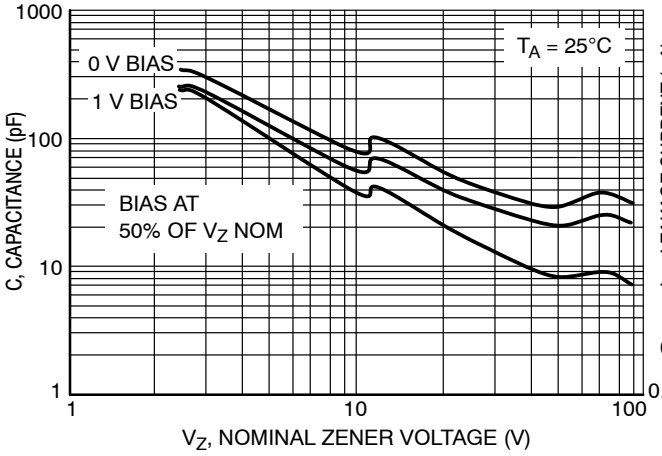



**Figure 5. Typical Capacitance**

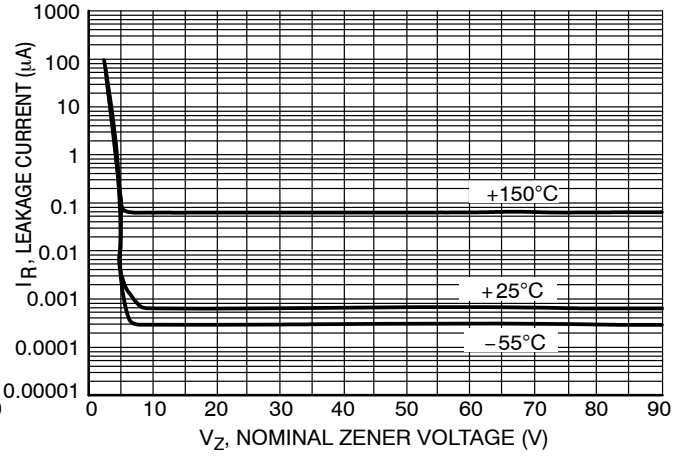



**Figure 6. Typical Leakage Current**

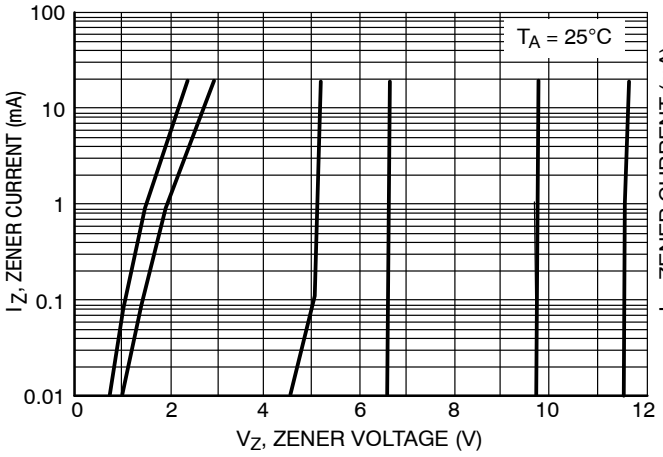



**Figure 7. Zener Voltage versus Zener Current ($V_Z$ Up to 12 V)**

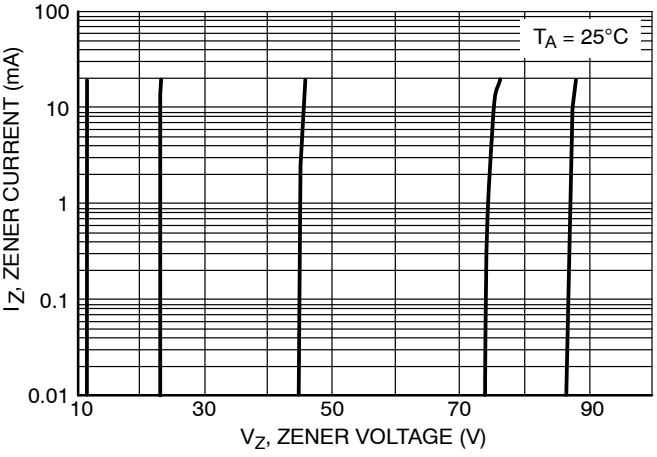



**Figure 8. Zener Voltage versus Zener Current (12 V to 91 V)**

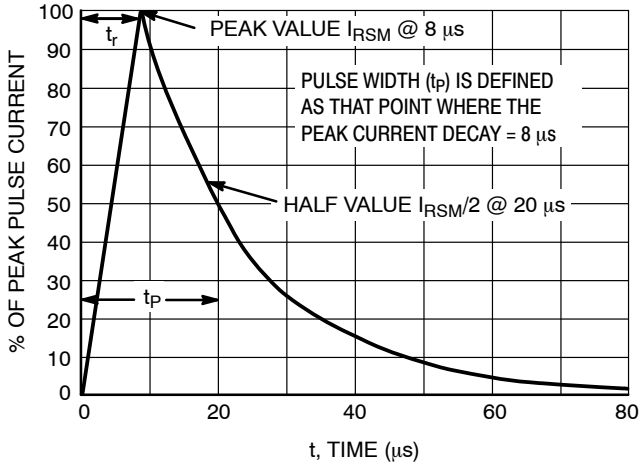
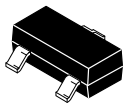



**Figure 9. 8 × 20 μs Pulse Waveform**



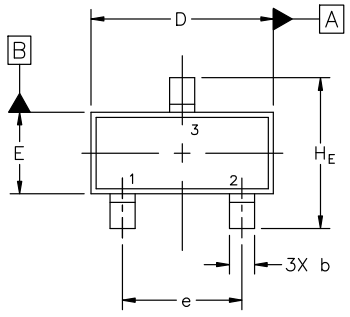
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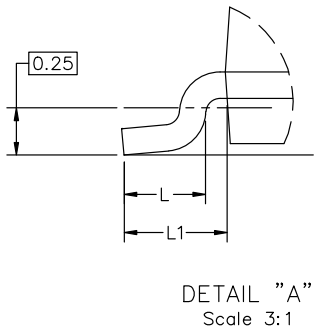
**SCALE 4:1**

**SOT−23 (TO−236) 2.90x1.30x1.00 1.90P**
CASE 318
ISSUE AU

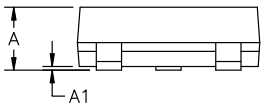
DATE 14 AUG 2024



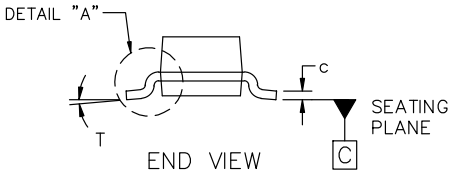

TOP VIEW




DETAIL "A"
Scale 3:1




SIDE VIEW

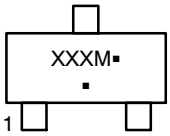



END VIEW

| MILLIMETERS | | | |
|---|---|---|---|
| DIM | MIN | NOM | MAX |
| A | 0.89 | 1.00 | 1.11 |
| A1 | 0.01 | 0.06 | 0.10 |
| b | 0.37 | 0.44 | 0.50 |
| c | 0.08 | 0.14 | 0.20 |
| D | 2.80 | 2.90 | 3.04 |
| E | 1.20 | 1.30 | 1.40 |
| e | 1.78 | 1.90 | 2.04 |
| L | 0.30 | 0.43 | 0.55 |
| L1 | 0.35 | 0.54 | 0.69 |
| $H_E$ | 2.10 | 2.40 | 2.64 |
| T | 0° | −−− | 10° |

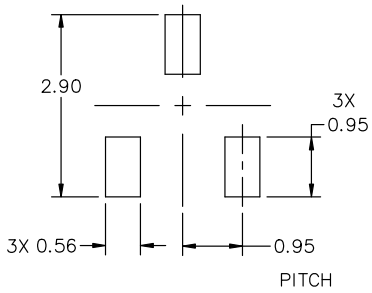
NOTES:
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 2018.
2. CONTROLLING DIMENSIONS: MILLIMETERS.
3. MAXIMUM LEAD THICKNESS INCLUDES LEAD FINISH. MINIMUM LEAD THICKNESS IS THE MINIMUM THICKNESS OF THE BASE MATERIAL.
4. DIMENSIONS D AND E DO NOT INCLUDE MOLD FLASH, PROTRUSIONS, OR GATE BURRS.

**GENERIC**
**MARKING DIAGRAM***




XXX = Specific Device Code
M = Date Code
▪ = Pb−Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, "G" or microdot "▪", may or may not be present. Some products may not follow the Generic Marking.




RECOMMENDED
MOUNTING FOOTPRINT

* For additional information on our Pb−Free strategy and soldering details, please download the onsemi Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

**STYLES ON PAGE 2**



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**SOT−23 (TO−236) 2.90x1.30x1.00 1.90P**
CASE 318
ISSUE AU

DATE 14 AUG 2024

STYLE 1 THRU 5:
CANCELLED

STYLE 6:
PIN 1. BASE
2. EMITTER
3. COLLECTOR

STYLE 7:
PIN 1. EMITTER
2. BASE
3. COLLECTOR

STYLE 8:
PIN 1. ANODE
2. NO CONNECTION
3. CATHODE

STYLE 9:
PIN 1. ANODE
2. ANODE
3. CATHODE

STYLE 10:
PIN 1. DRAIN
2. SOURCE
3. GATE

STYLE 11:
PIN 1. ANODE
2. CATHODE
3. CATHODE−ANODE

STYLE 12:
PIN 1. CATHODE
2. CATHODE
3. ANODE

STYLE 13:
PIN 1. SOURCE
2. DRAIN
3. GATE

STYLE 14:
PIN 1. CATHODE
2. GATE
3. ANODE

STYLE 15:
PIN 1. GATE
2. CATHODE
3. ANODE

STYLE 16:
PIN 1. ANODE
2. CATHODE
3. CATHODE

STYLE 17:
PIN 1. NO CONNECTION
2. ANODE
3. CATHODE

STYLE 18:
PIN 1. NO CONNECTION
2. CATHODE
3. ANODE

STYLE 19:
PIN 1. CATHODE
2. ANODE
3. CATHODE−ANODE

STYLE 20:
PIN 1. CATHODE
2. ANODE
3. GATE

STYLE 21:
PIN 1. GATE
2. SOURCE
3. DRAIN

STYLE 22:
PIN 1. RETURN
2. OUTPUT
3. INPUT

STYLE 23:
PIN 1. ANODE
2. ANODE
3. CATHODE

STYLE 24:
PIN 1. GATE
2. DRAIN
3. SOURCE

STYLE 25:
PIN 1. ANODE
2. CATHODE
3. GATE

STYLE 26:
PIN 1. CATHODE
2. ANODE
3. NO CONNECTION

STYLE 27:
PIN 1. CATHODE
2. CATHODE
3. CATHODE

STYLE 28:
PIN 1. ANODE
2. ANODE
3. ANODE




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